Week 1 – Tutorial

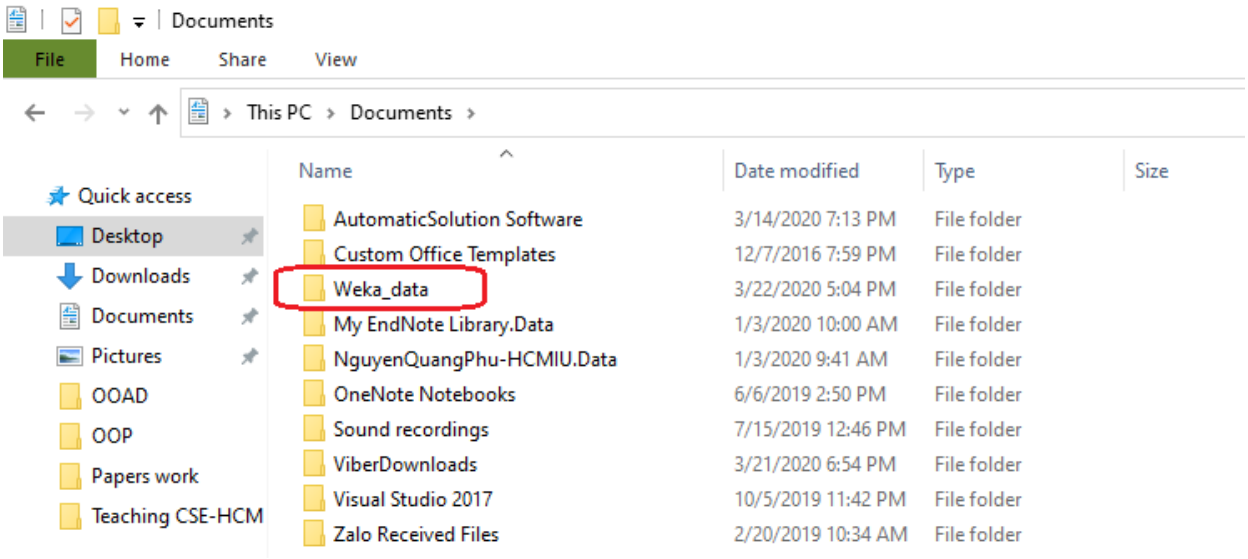
Class 1: Introduction to Weka

Download software: https://www.cs.waikato.ac.nz/ml/weka/

Please use the latest version: Weka 3.9.6

Install and open the Weka (see clip https://www.youtube.com/watch?v=nHm8otvMVTs - from 1:12-3:10)

How to copy data in Program file to your document: 3:30 to 4:21



Tutorial video:

1. Exploring the Explorer: https://web.microsoftstream.com/video/d3a1b1c0-15c5-4ac2-99be-ce409ea91f95
2. Exploring datasets: https://web.microsoftstream.com/video/047eaa16-37b1-481a-a949-60c4f09105b0
3. Building a classifier: https://web.microsoftstream.com/video/33755336-8560-4450-b0ee-03eafafb3b55
4. Using a filter https://web.microsoftstream.com/video/6387553d-862a-488e-a80f-bfe1db886e03
5. Visualizing your data: https://web.microsoftstream.com/video/10d75d9e-8220-4fbc-a4cd-fe5c3c9da53e

1.1. Introduction

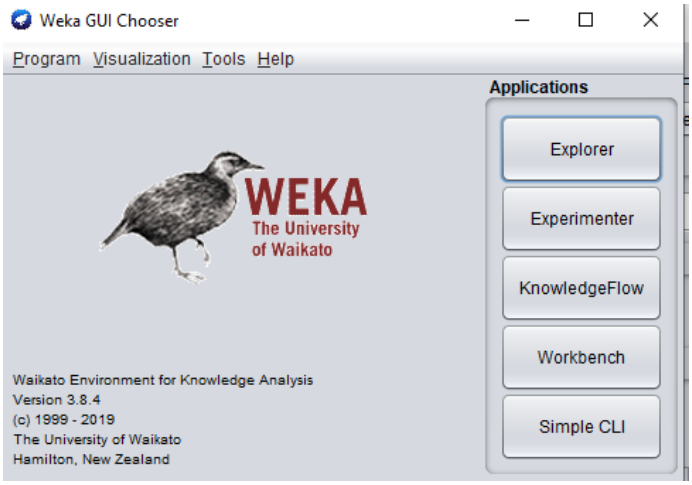
Weka is an open-source software available at www.cs.waikato.ac.nz/ml/weka. Weka stands for the Waikato Environment for Knowledge Analysis. It offers clean, spare implementation of the simplest techniques, designed to aid understanding of the data mining techniques. It also provides a work-bench that includes full, working, state-of-the-art implementations of many popular learning schemes that can be used for practical

data mining or for research.

In the first class, we are going to get started with Weka: exploring the "Explorer" interface, exploring some datasets, building a classifier, using filters, and visualizing your dataset. (See the lecture of class 1 by Ian H. Witten, [1])

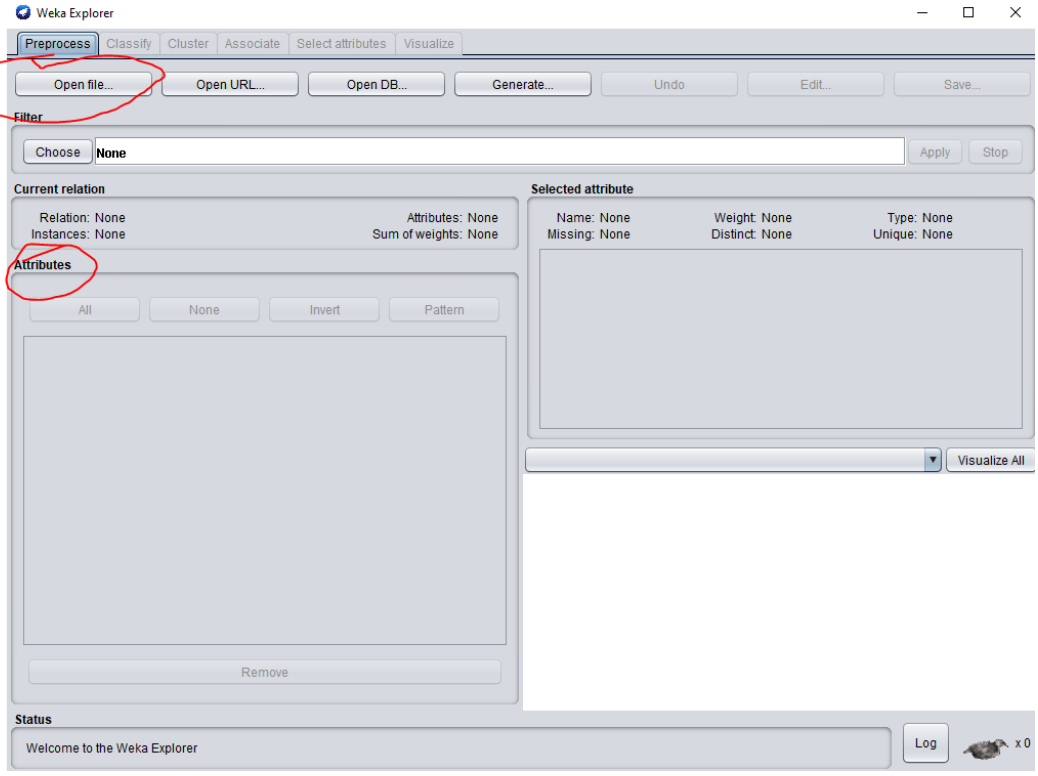
Task: Taking notes how you find the Explorer, and answering questions in the following sections

Open Weka



1. Explorer:
2. Experimenter: Performance comparisons
3. KnowledgeFlow: Graphical interface
4. Workbench (not in Weka 3.6.4)
5. Simple CLI: command-line interface
and click Explorer to continue 1.2

1.2. Exploring the Explorer





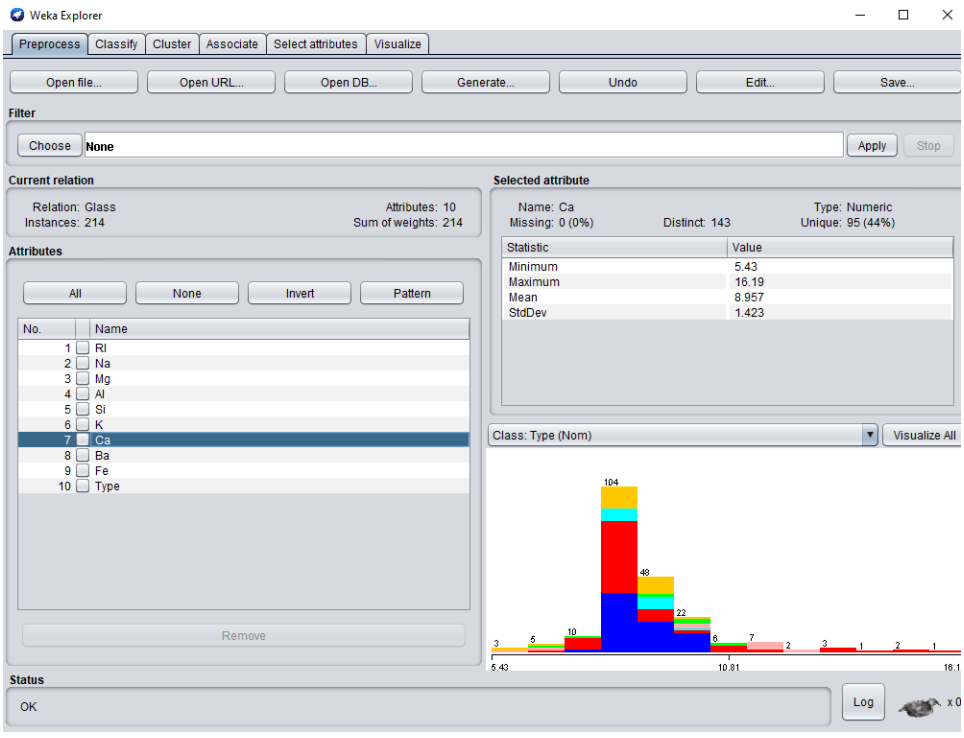
Introduce to
Preprocess
Classify panner
Cluster
Associate
Select attributes
Visualize
Practice: open the weather.nominal.arff

1.3. Exploring datasets

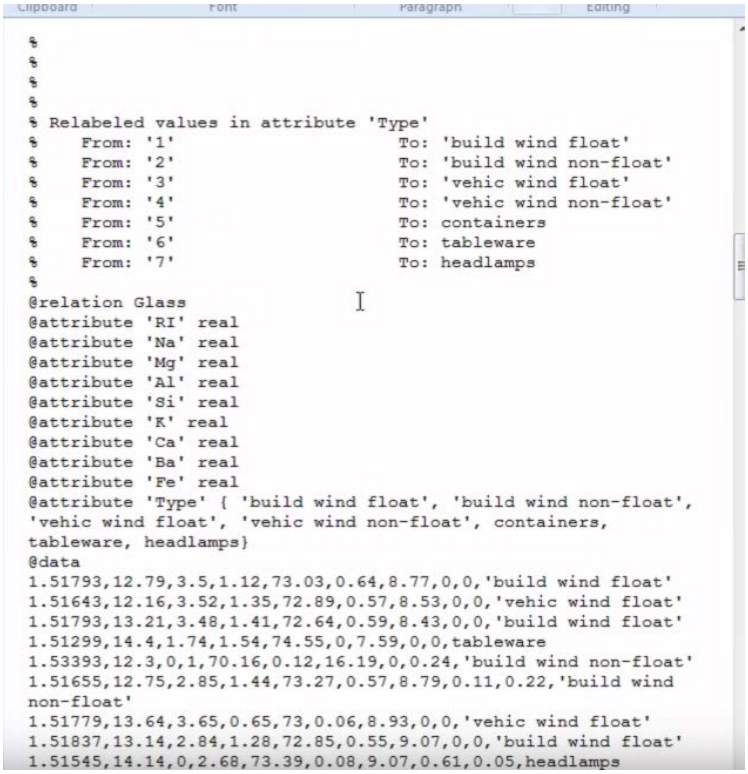
In dataset weather.nominal.arff, how many attributes are there in the relation? What are their values? What is the class and its values? Counting instances for each attribute value.

Dataset	Attributes	Values	#Instances
Relation: #Instances: #Attributes:			

Similarly, examine datasets: *weather.numeric.arff* and *glass.arff*



-Open glass.arff file using notepad to understand the struct of this file such as relation glass, attribute, ….



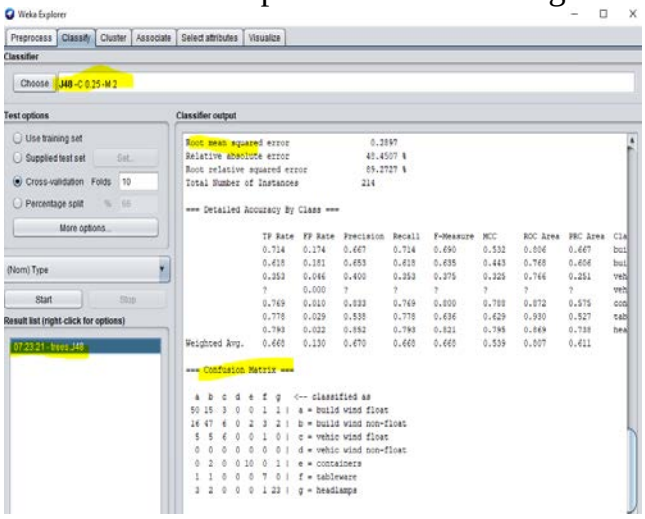
```
%
%
%
%
% Relabeled values in attribute 'Type'
%     From: '1'                    To: 'build wind float'
%     From: '2'                    To: 'build wind non-float'
%     From: '3'                    To: 'vehic wind float'
%     From: '4'                    To: 'vehic wind non-float'
%     From: '5'                    To: containers
%     From: '6'                    To: tableware
%     From: '7'                    To: headlamps
%
@relation Glass
@attribute 'RI' real
@attribute 'Na' real
@attribute 'Mg' real
@attribute 'Al' real
@attribute 'Si' real
@attribute 'K' real
@attribute 'Ca' real
@attribute 'Ba' real
@attribute 'Fe' real
@attribute 'Type' { 'build wind float', 'build wind non-float',
'vehic wind float', 'vehic wind non-float', containers,
tableware, headlamps}
@data
1.51793,12.79,3.5,1.12,73.03,0.64,8.77,0,0,'build wind float'
1.51643,12.16,3.52,1.35,72.89,0.57,8.53,0,0,'vehic wind float'
1.51793,13.21,3.48,1.41,72.64,0.59,8.43,0,0,'build wind float'
1.51299,14.4,1.74,1.54,74.55,0,7.59,0,0,tableware
1.53393,12.3,0,1,70.16,0.12,16.19,0,0.24,'build wind non-float'
1.51655,12.75,2.85,1.44,73.27,0.57,8.79,0.11,0.22,'build wind
non-float'
1.51779,13.64,3.65,0.65,73,0.06,8.93,0,0,'vehic wind float'
1.51837,13.14,2.84,1.28,72.85,0.55,9.07,0,0,'build wind float'
1.51545,14.14,0,2.68,73.39,0.08,9.07,0.61,0.05,headlamps
```

(7’40)

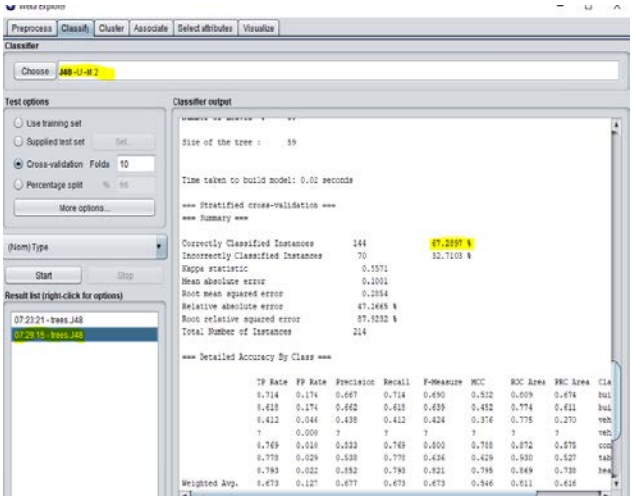
Create a file of ARFF format and examine it.

1.4. Building a classifier

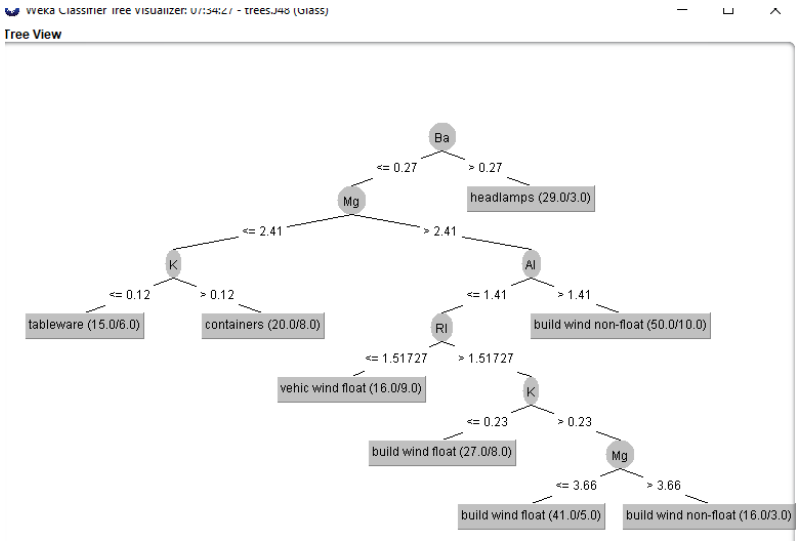
Examine the output of J48 to dataset glass.arff after applied start with pruned/unpruned



(a) Pruned



(b) unpruned



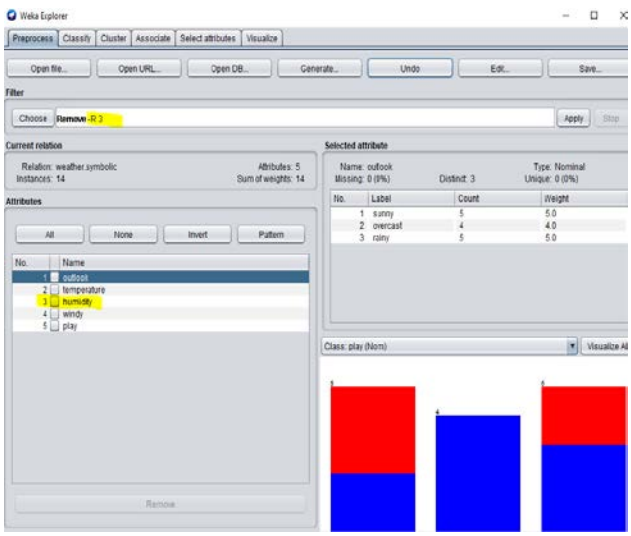
Please explain something about this figure

Algorithm	Pruned/unpruned	minNumObj	Leaf size	Correctly Classified Instances

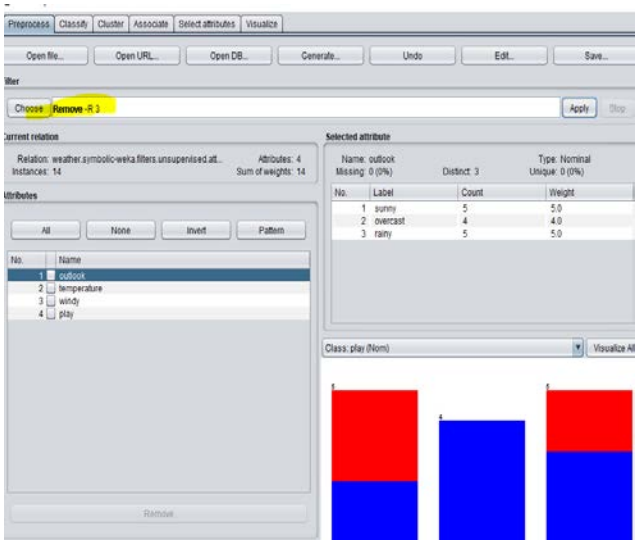
Evaluate the confusion matrix every time running an algorithm…

1.5. Using a filter

Follow the instructions in [1], and remark



Before



After

_Use a filter to remove an attribute □

- What are attributeIndices? -

_Remove instances where humidity is high □

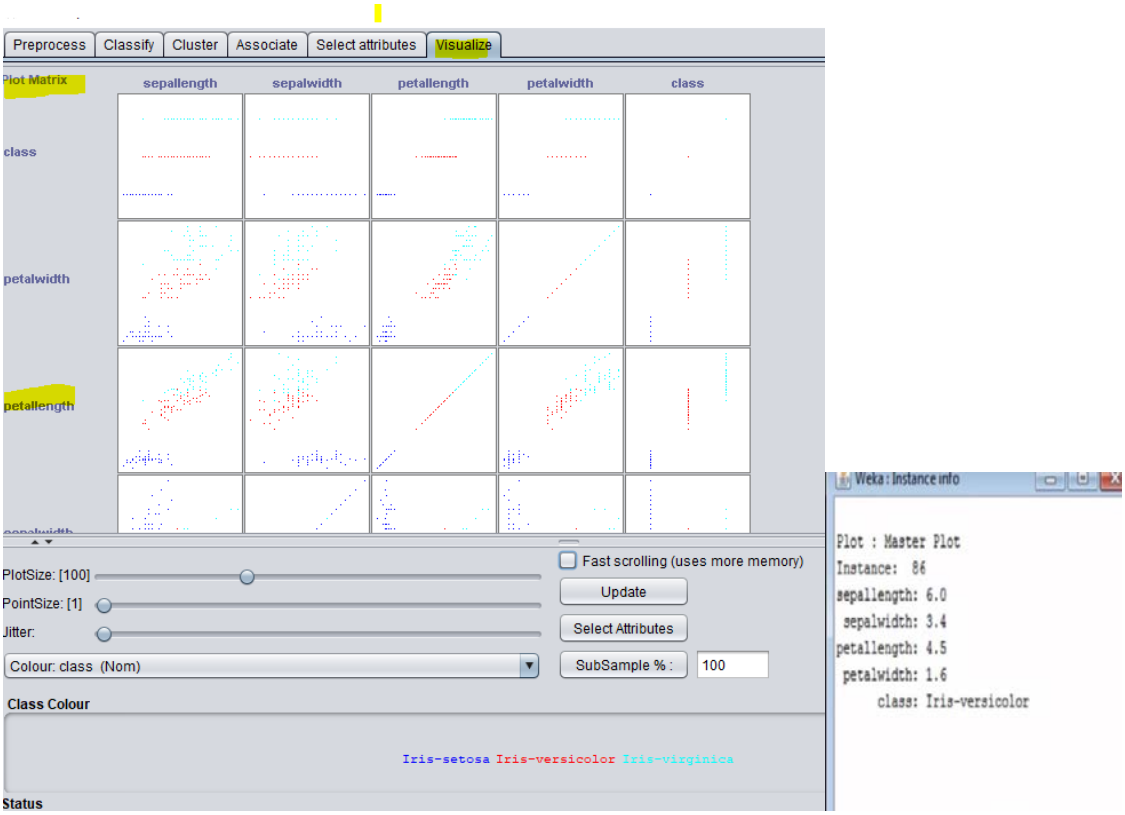
- What are nominalIndices? -

_Fewer attributes, better classification:

Follow the instructions in [1], review the outputs of J48 applied to glass.arff:

Filter	**Leaf size**	**Correctly Classified**	**Remark**
Original			
Remove Fe			
Remove all attributes except RI and MG			

1.6. Visualizing your data



Follow the instructions in [1], how do you find "Visualize classifier errors"? After running **J48** for *iris.arff*, determine:

- How many instances are predicted wrong? -
- What are they?

Instance	Predicted class	Actual class